

```
\import java.util.Scanner;
```

```
class StackArray {
```

```
    private int[] arr;
```

```
    private int top;
```

```
    private int capacity;
```

```
    public StackArray(int capacity) {
```

```
        this.capacity = capacity;
```

```
        arr = new int[capacity];
```

```
        top = -1; // Initially empty stack
```

```
    }
```

```
    public boolean push(int element) {
```

```
        if (isFull()) {
```

```
            System.out.println("Stack is full!");
```

```
            return false;
```

```
        }
```

```
        top++;
```

```
        arr[top] = element;
```

```
        return true;
```

```
    }
```

```
    public int pop() {
```

```
        if (isEmpty()) {
```

```
            System.out.println("Stack is empty!");
```

```
            return -1; // Can modify to throw an exception
```

```
        }
```

```
        int element = arr[top];
```

```
        top--;
```

```
        return element;
```

```
    }
```

```
    public int peek() {
```

```
    if (isEmpty()) {  
        System.out.println("Stack is empty!");  
        return -1; // Can modify to throw an exception  
    }  
    return arr[top];  
}
```

```
public boolean isEmpty() {  
    return top == -1;  
}
```

```
public boolean isFull() {  
    return top == capacity - 1;  
}
```

```
public void displayStack() {  
    System.out.print("[");  
    for (int i = 0; i <= top; i++) {  
        System.out.print(arr[i] + ", ");  
    }  
    System.out.print("]");  
}  
}
```

```
public class StackDriver {  
  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
  
        System.out.print("Enter stack capacity: ");  
        int capacity = scanner.nextInt();  
  
        StackArray stack = new StackArray(capacity);  
  
        int choice;
```

```
do {
    System.out.println("\nStack Operations:");
    System.out.println("1. Push");
    System.out.println("2. Pop");
    System.out.println("3. Peek");
    System.out.println("4. Display Stack");
    System.out.println("5. Exit");

    System.out.print("Enter your choice: ");
    choice = scanner.nextInt();

    switch (choice) {
        case 1:
            System.out.print("Enter element to push: ");
            int element = scanner.nextInt();
            stack.push(element);
            break;
        case 2:
            int poppedElement = stack.pop();
            if (poppedElement != -1) {
                System.out.println("Popped element: " + poppedElement);
            }
            break;
        case 3:
            int topElement = stack.peek();
            if (topElement != -1) {
                System.out.println("Top element: " + topElement);
            }
            break;
        case 4:
            stack.displayStack();
            break;
        case 5:
            break;
        default:
```

```
        System.out.println("Invalid choice!");
        break;
    }
} while (choice != 5);

scanner.close();
System.out.println("Program terminated.");
}
}
```